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LEAN MANUFACTURING, GREEN VALUE CO-CREATION, AND GREEN PRODUCT INNOVATION: ESTABLISHING CONCEPTUAL LINKAGES

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ABSTRACT:

Collaborative operational interventions have led to great support for long-term competitiveness and sustainable innovation in the context of the growing importance of ecological sustainability. This study builds a relationship between lean manufacturing (LM), green value co-creation (GVC) and green product innovation (GPI). This nexus is a mechanism to describe how these factors work together to achieve sustainability-oriented organizational performance. The study adopted a qualitative research approach, engaging a mix of (i) content analysis, (ii) thematic analysis, and (iii) conceptual framework analysis of secondary data. The secondary data was derived from academic publications, including peer-reviewed articles, books, chapters, and even conference proceedings. The findings of the paper advocate that lean manufacturing enables resource efficiency, promotes ecologically balanced sustainable production interventions, and minimizes waste, which subsequently influences green product innovation. Also, GVCC inculcates the spirit of knowledge sharing, environmental participation, and stakeholder collaboration amongst all supply chain players, thereby permitting sustainable product development. Integration of lean operational practices with green strategies helps in realization of stronger innovation capability, improved eco-performance, and assurance of long-term sustainable outcomes. This enables sustainable competitiveness through technologically driven lean operational practices. The proposed conceptual framework signifies the nexus between LM, GVCC and GPI for achieving sustainable competitive advantage. Futuristic researchers can empirically delve towards validating and establishing the proposed relationships in various organizational and industry contexts.

Keywords: *Lean Manufacturing, Green Product Innovation, Green Value Co-Creation, Sustainability.*

1. Introduction

Businesses are facing immense pressure to address eco-centric sustainability challenges by improving their operational efficiency. Change in customer expectations and increasing eco concerns have propelled organizations to opt for innovative and sustainable manufacturing practices (Wolf, 2014). In this light, the concept of lean manufacturing (LM) integrates process efficiency, optimized resource utilization, and waste reduction in production systems. In eco-centric decision-making and ecologically-oriented innovative designing, the gamut of supply chain players, including customers, suppliers, and all the stakeholders, is being engaged for developing collaborative sustainability practices (Yousuf, 2021). Green product innovation (GPI) has received paramount attention in contemporary times owing to its impetus on sustainable business performance by minimizing the detrimental eco-impacts. Researchers indicate that long-term competitiveness and innovation capability are improved via stakeholder pressure, green capabilities, and through collaborative eco-practices (Singh et al. 2022; Chen and Chang, 2013). Previous research has studied operational sustainability, collaborative environmental interventions, practices, and green innovation in isolation. No conclusive study has been found to explore these concepts synergistically contributing to sustainability-centric organizational performance. This was the identified gap. Based on this gap, the objective of the study is to qualitatively examine the association between LM, GVCC, and GPI by engaging a combination of qualitative research approaches with content analysis, thematic analysis, and conceptual framework analysis.

2. Literature Review

2.1 Overview of LM, GVCC, and GPI

Within the ambit of production systems, Lean Manufacturing lays impetus on improving operational efficiency via optimized resource utilization and waste reduction. A plethora of benefits is offered by lean practices like JIT and Kaizen, including (a) increased customer value, (b) better operational performance, and (c) escalated product quality metrics (Womack & Jones, 1997; Shah & Ward, 2003). Lean Manufacturing has been found to be pivotal to ecocentric sustainability as it reduces energy consumption and promotes optimal material usage (King & Lenox, 2001). GVCC is referred to as the total of collaborative efforts by all the players of the supply chain for generating both social and environmental value. It lays emphasis on engaging in sustainable interventions and ecologically oriented product development. Previous studies have narrated green collaboration as a generator of environmental performance, customer satisfaction, innovation capability, and trust (Vargo and Lusch, 2008; Prahalad and Ramaswamy, 2004; Chen, 2010).

GPI encompasses the development of eco-centric products through:

a) Usage of recyclable materials,

b) Engagement of cleaner technologies,

c) Eco-designing,

d) Employment of energy-efficient processes.

Research reveals that green innovation:

a) Escalates organizational performance,

b) Establishes compliance with eco-regulations,

c) Improves competitive advantage (Dangelico & Pujari, 2010; Chen et al., 2006).

Evidences from previous researches indicate that LM and GVC positively influence GPI via eco-centric collaboration and through operational efficiency (Aboelmaged, 2018).

2.2 Lean Manufacturing and GPI.

Augmentation of GPI is facilitated by lean manufacturing through:

a) Waste minimization regimes,

b) Effective utilization of sustainable resources, and

c) Increased process efficiency.

An ambience for sustainable innovation is supported by LM via judicious energy consumption and lowered material waste (Islam, 2024). Eco-design practices that minimize the detrimental impact of products across their life cycle are encouraged by lean systems (Kouser et al., 2025). Both green innovation and LM work in tandem to advocate cleaner production technologies, sustainability, and optimized resource consumption (Levi- Bliech, and Dehan, 2025). Previous research has pointed LM to be dedicated towards allotment of resources for greener technologies and paving the way towards innovative product development (Hou, 2026). Mediated via sustainable operational practices, lean production systems herald positive influence on environmental performance and social sustainability (Afum et al., 2023). Subsequently, organizations adhering to an amalgamated approach of GI with LM tend to achieve two-fold benefits of:

1. Attainment of long-term sustainability goals and
2. Enhanced economic and environmental performance of organizations (Altarazi, 2025).

2.3 GVCC and GPI

GVCC is a collaborative process between companies, customers, suppliers, regulators and other stakeholders, which enables green product innovation. This collaboration facilitates the organizations to become more environmentally aware and create an eco-friendly product that reflects market expectations (Putri et al., 2025). Research suggests that stakeholder engagement can lead to greater environmental knowledge sharing, innovation capacity, and sustainable business outcomes (Wang & Ahmad, 2024). Open innovation and collaborative practices also promote sustainable solutions and enhanced environmental performance (Padthar & Ketkaew, 2024). In addition, research shows that green collaborative innovation involves technology, stakeholder engagement, and knowledge fusion to facilitate sustainable product development (Huang & Chen, 2025). Similarly, successful low-carbon innovation ecosystems require value co-creation between companies and stakeholders to ensure the sustainability in the long term and competitiveness of these ecosystems (Zhang et al., 2025; Tao, 2026). As such, GVCC is an important linkage between stakeholder cooperation and successful green product innovation.

2.4 LM and GVCC

LM and GVCC share a strong connection, as both share a focus on efficiency, sustainability, and continuous improvement. The main benefits of lean manufacturing are waste reduction, energy savings and improved operational performance, which all help to achieve environmentally responsible business practices (Younnes, 2023). However, lean systems also enhance the cooperation among supply chain partners and stakeholders, which further aids GVCC activities (Olaleye et al., 2025). Lean and green approaches also foster innovation, knowledge sharing, and sustainable product development, increase stakeholder satisfaction and long-term competitiveness (Benabdellah, 2024; Duarte et al., 2025; Bonamigo et al., 2024; Gál, 2025; Pierli et al., 2026).

2.5 Nexus between LM, GVCC, GPI

In the last few years, the literature has focused on the relationship between LM, GVCC and GPI. Lean manufacturing can help to optimize resource usage, reduce waste, and enhance ecological sustainability by streamlining production processes (Pierli et al., 2026; Gál, 2025). GVCC deepens this relationship by fostering collaboration and knowledge sharing among firms, customers, suppliers and stakeholders. This collaboration can help them better understand the environmental needs and create sustainable solutions more effectively (Siemieniako et al., 2025; Bonamigo et al., 2024). In addition, lean manufacturing has been noted as a key direct enabler of GPI, as operational systems that are efficient enable firms to concentrate more effectively on eco-friendly product innovation. At the same time, green value co-creates innovation ability and sustainable performance through better stakeholder engagement and environmental collaboration (Sulaiman, 2025; Kouser et al., 2025). In general, these factors are well interrelated, and their combined impact can enhance the sustainability, competitiveness and future business performance of organizations. This is why the present study aims to develop a conceptual framework to analyse the relationships among LM, GVCC, GPI, and sustainability-oriented organisational performance. The framework explains how lean manufacturing helps address operational efficiency, waste minimisation, and sustainability awareness issues, and how GVCC is a value creation process designed to increase collaboration, engagement, and knowledge sharing between companies, customers, suppliers, and other actors. In combination, these factors contribute to green product innovation for long-term organisational sustainability and competitive advantage.

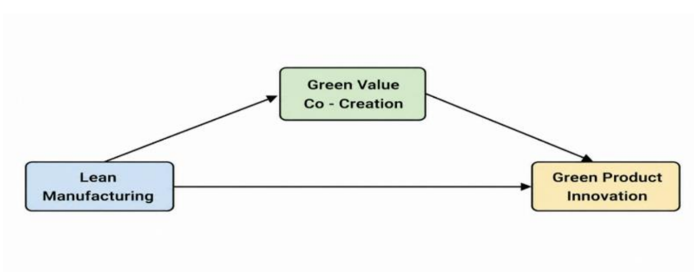

Figure 1: Conceptual Framework Proposed by Author

The framework suggests that LM has a direct impact on GPI, and it also supports GVCC between firms and stakeholders. Moreover, GVCC also has a constructive impact on GPI, linking to collaboration, sustainable practices and the sharing of environmental knowledge. In addition, GPI contributes to the improvement of the sustainability-oriented organizational performance through the benefits of enhancing the ecological outcome, innovation capability, and competitiveness of the firm in the long-term. The arrows in the framework indicate conceptually recognized relationships that were identified by thematic and literature analysis, but not statistically tested causal relationships. Therefore, the framework emphasizes the interrelatedness between lean operational practices and collaborative environmental strategies towards sustainability and competitive advantage.

3. Research Methodology

A qualitative approach has been utilized in the present research, which uses content analysis, thematic analysis, and conceptual framework analysis (CFA) to examine the association between LM, GVCC and GPI. According to Creswell (2017), qualitative research helps in understanding concepts and theoretical relationships in management research. The research primarily utilises secondary sources of data from peer-reviewed journals, books, and academic publications related to lean manufacturing, value creation, sustainability and green product innovation. Relevant and recent studies were included, while duplicate and irrelevant sources were excluded from the review process. Krippendorff (2018) explained that content analysis is used to identify recurring patterns and meanings within textual information. Therefore, content analysis was applied to examine recurring findings related to lean and green practices. Braun and Clarke (2006) stated that thematic analysis helps in identifying and analyzing important themes within qualitative data. Accordingly, thematic analysis was used to classify the literature into themes related to LM, GVCC, and GPI. Furthermore, conceptual framework analysis was employed to explain the relationships among the selected variables. Jabareen (2009) suggested that conceptual framework analysis helps in developing theoretical relationships among concepts in qualitative studies. Hence, a conceptual framework was developed in this study to explain the role of LM and GVCC in the improvement of GPI and green value-oriented organisational performance.

4. Findings

The findings are derived from qualitative interpretation, thematic analysis, and conceptual framework analysis of the reviewed literature. Several recurring themes and emerging insights associated with sustainable business practices and innovation-oriented organizational strategies were identified.

- The study reveals that lean manufacturing indirectly strengthens green product innovation by creating operational conditions that support stakeholder-driven green collaboration.
- GVCC was identified as a knowledge integration mechanism through which businesses transform environmental collaboration into sustainable product development outcomes.
- The findings indicate that waste reduction practices in lean systems contribute not only to efficiency but also to the development of eco-design capabilities and sustainable innovation culture within firms.
- The study highlights that firms adopting isolated green innovation practices may achieve limited outcomes unless supported by collaborative stakeholder participation and lean operational structures.
- Conceptual framework analysis shows that GVCC strengthens the strategic linkage between operational sustainability and market-oriented green innovation.
- The research suggests that sustainable competitiveness increasingly depends on the integration of internal process efficiency and external environmental collaboration rather than relying on technological innovation alone.

5. Discussion

The research demonstrates how LM and GVCC are able to demonstrate a synergic effect on the sustainable innovation and business performance of an organization. The literature review and findings show that lean practice can lead to waste reduction, better use of resources and the development of efficient production systems, which enable environmentally friendly product development. While reducing costs remains a primary focus in lean systems, today there is an increased emphasis on supporting sustainability goals and organizational improvement over the long term (Pierli et al., 2026). The effects of green product innovation are also improved when organisations engage customers, suppliers and stakeholders in environmental activities, the study finds. GVCC facilitates the sharing of ideas, environmental knowledge, and sustainable solutions and contributes to the development of products that align with market and environmental expectations. It is found that collaboration has increasingly become more than a mere business activity and is an important aspect of sustainable innovation (Huang & Chen, 2025). Another key finding of the study is that sustainable competitiveness can't be realized by technology alone. Additionally, the ability to cooperate with external parties and improve the efficiency of internal operations is important for organizations to build their innovation capability and environmental performance. The conceptual framework can be used to understand how the use of lean operational practices and collaboration with stakeholders for environmental issues work together to enhance the organization's performance towards sustainability.

6. Conclusion

The study shows the interconnection of three identified study variables (LM, GVCC, GPI) in the sustainable organization development process. Elements of lean manufacturing (operational efficiency, resource management, etc.) and green value co-creation (increased stakeholder participation, environmental collaboration, etc.) are combined. These all help to establish sustainable product and business performance in the long term. The study also reveals that sustainability and competitive advantage are only possible with technological progress, not just. Operational efficiency and working together on environmental strategies must be treated equally. The conceptual framework proposed here makes it possible to understand the link between lean systems and green collaboration with sustainable innovation and organization sustainability. The findings of the study also suggest that future researchers could apply the empirical and quantitative approaches for validation of the proposed relationships in other industries and organizational context for further practical validation.

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